GG-S-590B
October 21, 1976
SUPERSEDING
Int. Fed. Spec. GG-S-00590a(DSA-DM)
August 12, 1969 and
Fed. Spec. GG-S-590
October 13, 1961
(See 6.3)

FEDERAL SPECIFICATION

SPATULA, DENTAL, CEMENT

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers dental spatulas suitable for mixing cements.

1.2 Classification. Spatulas covered by this specification shall be of one type, and in the following styles, as specified (see 6.1):

Style 313.
Style 324.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-585	- Boxes, Wood, Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123	- Marking for Shipment (Civil Agencies).

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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Military Specifications:

MIL-P-116	- Methods of Preservation-packaging.
MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water-vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The spatulas shall be in accordance with the applicable requirements of GG-I-526, with the options and the additional requirements specified herein.

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3.1.1 Figures. The figures forming a part of this specification are not production drawings. Figures establish dimensional limitations and control design and configuration. The supplier shall be responsible for preparing his own shop drawings. The supplier shall select dimensions and tolerances within the limitations prescribed on the figures to provide a product with proper design, configuration and function. No deviations from the dimensional limitations prescribed on the figures are permissible without prior approval of the contracting officer.

3.1.2 Standard sample. The finish and workmanship shall be equal to or better than the standard sample (see 3.6, 3 8 and 6.2).

3.2 Material. Spatulas shall conform to the requirements of class 4, type 420 or type 440A stainless steel.

3.3 Hardness. After proper stress relieving, the Rockwell hardness of the operative end (blade) shall be C40 to C50 for instruments fabricated from type 420 steel, and C52 to C56 for instruments fabricated from type 440A steel. Handle and shank shall be properly tempered and shall have a hardness suitable for the purpose intended.

3.4 Style, design and dimensions. The spatulas shall be single-ended instruments of one-piece construction. Spatulas shall be suitable for mixing cements. The blade of the spatula shall be properly tapered to provide the required flexibility in mixing cements. The style, design and dimensions of the spatulas shall conform to figure 1 for style 313 and figure 2 for style 324. Spatula handle shall be straight.

3.5 Handles. Spatulas handles shall be smooth, or shall have fillet or notch type serrations. For serrated handles the depth of the serrations shall be 0.001 to 0.004 inch. There shall be 40 ± 10 serrations per inch on seven faces and one smooth face for identification marking.

3.6 Finish. Smooth spatulas handles shall have a satin, mirror, or bright polished finish. Those instrument handles with fillet type or notch type serrations shall have a bright finish. The operative ends and the shanks of the spatulas shall have a mirror or a bright polished finish. The finish of the operative ends shall be equal to be or better than the standard sample (see 3.1.2 and 6.2).

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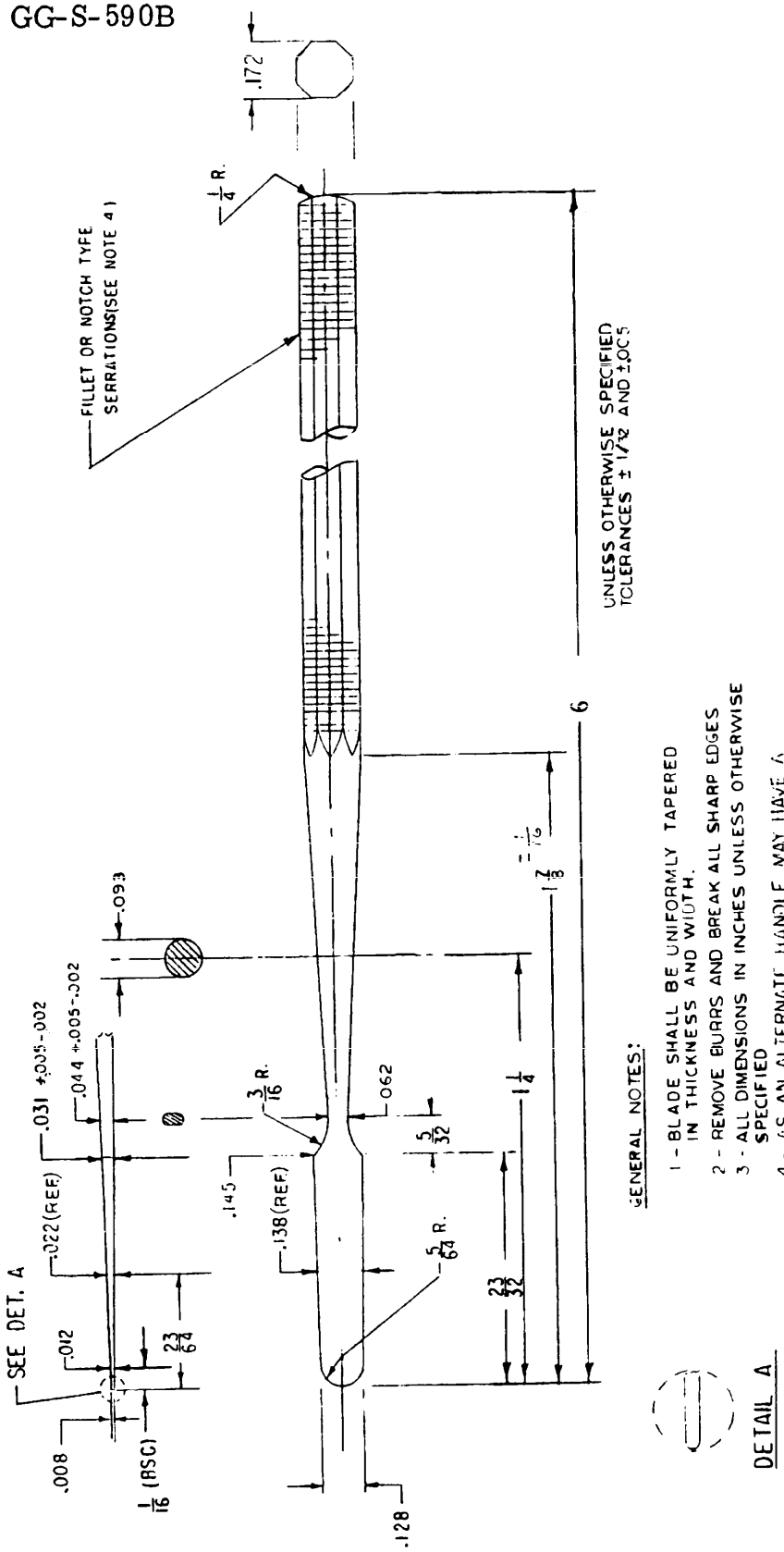


Figure 1 - Spatula, Dental, No. 313

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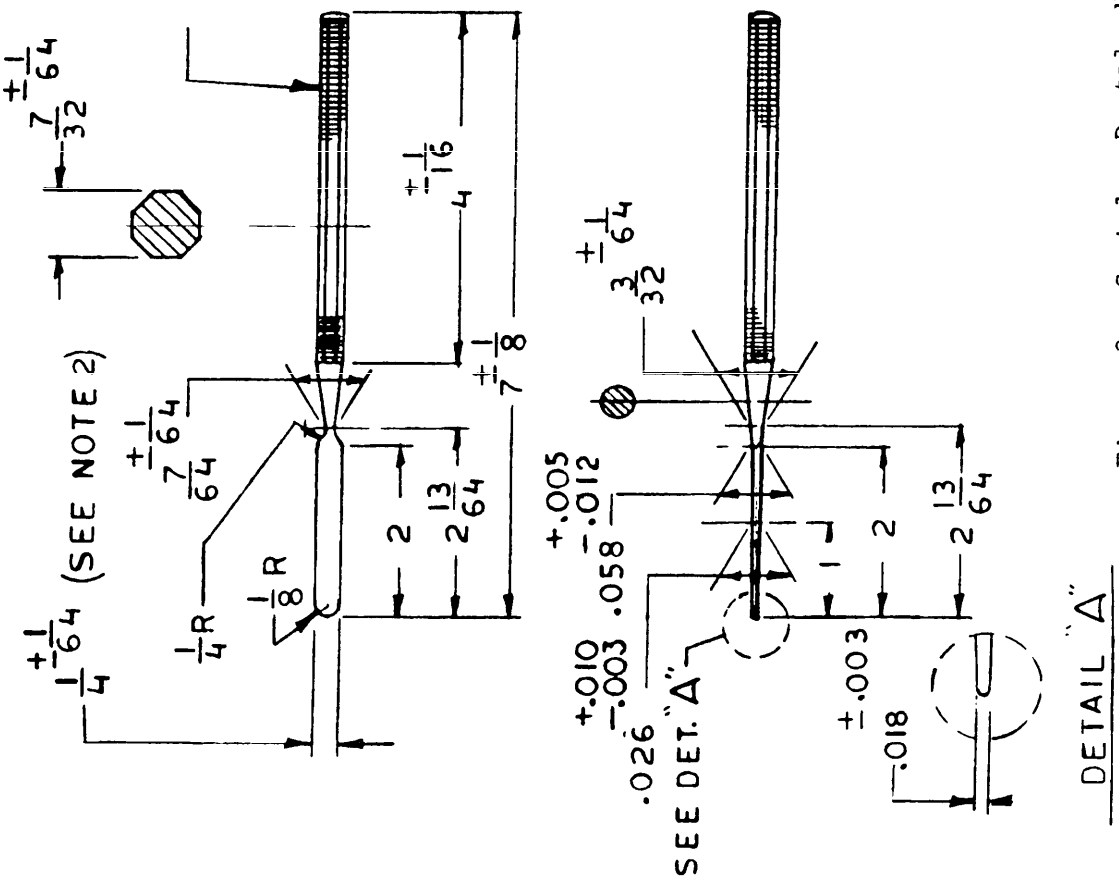


Figure 2 - Spatula, Dental, No. 324

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3.7 Identification marking. In addition to the marking requirements of GG-I-526, style 313 spatula shall be marked with number "313" and style 324 spatula shall be marked with number "324". All marking shall be legible and permanent. The word "stainless" shall also be marked on instruments of foreign manufacture. As an alternate to the word "stainless", the use of a registered tradename identifying the steel as stainless steel shall be acceptable, provided prior approval is received from the cognizant contracting officer (see 6.1).

3.8 Workmanship. In addition to the requirements of GG-I-526, the operative ends shall be free of discolorations and coarse grind marks. Workmanship shall be equal to or better than the standard sample (see 3.1.2 and 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the metal and packaging materials, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component or material, the date of testing, the test method, individual test results, the specification requirements, the lot number, the lot size and the sample size.

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4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties). Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Examination. Spatulas shall be examined for defects, including but not limited to those listed in table I.

TABLE I. Classification of defects.

Categories	Defects
Critical:	None defined
Major A:	
101	Style and design not as specified for the applicable style.
102	Finish on operative ends not as specified.
103	Operative ends not free of coarse grind marks, cracks, nicks, gouges, burrs or other imperfections.
104	Operative end bent.
Major B:	
151	Operative end dimensions not as specified.
Minor A:	
201	Serrations not as specified, or incomplete, as applicable.
202	Handle not straight.
203	Finish on handle and shank not as specified.
204	Instrument identification not present, not legible, not permanent, or improperly located.
205	Instrument not permanently and legibly marked with name or tradename of manufacturer.
206	Instruments not free from dirt, oil, polishing compound, or foreign matter.
Minor B;	
251	Dimensions (other than 151) not as specified.

The inspection levels and acceptable quality level (AQL) shall be as listed in table II. The unit of product for examination shall be one spatula.

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TABLE II. Examination Criteria

Type Defect	Inspection Level	AQL (Percent Defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Testing. Spatulas shall be tested as specified in table III and 4.3. Inspection level and acceptable quality level (AQL) shall be as listed in table III. The unit of product for testing shall be one spatula.

TABLE III. Test Criteria

Characteristic	Component	Requirement	Test procedure	Inspection level	AQL
Composition Analysis	-	3.2 and GG-I-526	GG-I-526	**	*
Hardness	Spatula	3.3 and GG-I-526	GG-I-526	S-2	1.0
Corrosion Resistance	Spatula	GG-I-526	GG-I-526	S-2	1.0

* Acceptance number shall be zero for all sample sizes.
** A sufficient amount from each lot or metal used in the construction of the spatulas shall be selected for chemical analysis.

4.2.3 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging and marking requirements of Section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 percent defective.

TABLE IV. Examination of preparation for delivery

Container	Not as specified.
Contents	Not as specified.
Marking	Omitted, incorrect, illegible, improper size, location, sequence or method of application.
Packaging and packing materials	Component missing or damaged.
Workmanship	Bulging of distortion of containers. Cushioning inadequate or improper.

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4.3 Inspections not listed in [illegible]

4.2.3.1 Examination of closure, waterproofing and banding of containers. When unit packages are required to comply with PPP-B-636, examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of that specification.

4.3 Tests. Tests shall be conducted to determine compliance with specification requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning, drying and preservative. Each spatula shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116. Preservative shall not be applied.

5.1.1.2 Unit packaging. Each spatula shall be wrapped in neutral tissue or other suitable anticorrosive material. Each spatula shall be further packaged in accordance with MIL-P-116, method IA-8 or method IC-1.

5.1.1.3 Intermediate packaging. Twelve (12) spatulas shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.2 Level C. Spatulas shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A. Two hundred eighty-eight (288) spatulas shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-585, class 3, style 3, PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with the appendix of the applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproofed in accordance with PPP-B-636.

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5.2.2 Level B. Two hundred eighty-eight (288) spatulas shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-585, class 2, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. Spatulas shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.3.2 Military activities. Each unit, intermediate package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Style required (see 1.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
d. Approval of tradename, if applicable (see 3 7).

6.2 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids or request for proposal (see 3.1.2, 3.6 and 3.8).

6.3 Supersession data. This specification includes the requirements of MIL-S-36793, dated September 9, 1971.

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